ROMAN NALAPKO

PYTHON BACKEND & APPLIED AI ENGINEER

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PROFESSIONAL SUMMARY

Python backend and applied AI engineer with 3+ years of project-based experience building RAG systems, asynchronous APIs, LLM integrations, and automation workflows. Develops testable services with FastAPI, PostgreSQL, SQLAlchemy, Qdrant, Docker, Cloudflare, and CI, from requirements through validation and handoff.

TECHNICAL SKILLS

Languages & Backend: Python, SQL, FastAPI, asyncio, REST APIs, SQLAlchemy, Pydantic, Alembic, webhooks

Applied AI: RAG, hybrid retrieval, embeddings, semantic search, LLM evaluation, prompt/context engineering, OpenAI-compatible APIs, Gemini API, Whisper

Data & Cloud: PostgreSQL, Qdrant, Cloudflare Workers, D1, KV, pandas, document processing

Quality & Delivery: pytest, Ruff, Docker, Docker Compose, GitHub Actions, structured logging, Prometheus, Linux, Git

EXPERIENCE

Independent Python Backend & Applied AI Engineer | *Private Client & Project-Based Work* | *Mar 2023 – Present* | *Remote*

- Delivered Python backend and automation work from requirements through API design, data modeling, validation, testing, Dockerized execution, and technical handoff.

Private Client Delivery — AI Personal Finance Assistant | *Python, PostgreSQL, Gemini, Whisper*

- Built an AI-assisted finance workflow with Telegram as the user interface, supporting both text and transcribed voice inputs while keeping the implementation and client identity private.
- Converted model responses into validated application data, checked amounts, currencies, categories, and required fields, and required explicit user confirmation before persistence.
- Designed PostgreSQL schemas and Alembic migrations with user-scoped queries, atomic multi-transaction writes, currency-separated summaries, and consistent rollback behavior.
- Added XLSX export, service health checks, automated parsing and persistence tests, Docker configuration, and handoff documentation for ongoing operation.

SELECTED ENGINEERING PROJECTS

AI RAG Platform | *Python, FastAPI, PostgreSQL, Qdrant, Docker* | GitHub

- Built a multi-tenant document intelligence backend with PDF/TXT ingestion, durable indexing jobs, versioned database migrations, and separate API and worker services.
- Implemented hybrid retrieval by combining Qdrant dense-vector search with PostgreSQL full-text search through Reciprocal Rank Fusion and tenant-scoped filters.
- Added bcrypt/JWT authentication, user-owned knowledge bases, conversation memory, and SSE answers that stream source citations with grounded responses.
- Created offline RAG evaluation, Prometheus metrics, structured JSON logs with request IDs, provider failure mapping, and compensating cleanup for partial indexing failures.
- Designed pluggable local and cloud OpenAI-compatible model integrations and added 118 automated tests covering API contracts, authentication, retrieval, and failure paths in CI.

EDUCATION & CREDENTIALS

Bachelor's Program in Information Systems and Technologies (F6) | Danube Institute of National University 'Odesa Maritime Academy' | Expected 2030

Selected certifications: Database Integration and Management — Microsoft; Linux Commands & Shell Scripting Essentials V2 — IBM; Trustworthy Generative AI — Vanderbilt University

LANGUAGES

Ukrainian — Native | Russian — Native | English — Intermediate (B1)